

# Event & Room Occupancy Management System

*Project Review 1 — Abstract & Design Proposal*

## TEAM

Sec 10 · Team 1

## TRIMESTER

2026–2027, Tri-4

## MEMBERS

J. Geya Reddy · Kamtam Priyanka

# Problem Statement

Students and administrators lack a single, reliable way to see which campus rooms are free and to manage events without last-minute chaos.



## Room availability is invisible

Students struggle to find free rooms during college hours — there is no shared, real-time view.



## Spot registrations disrupt planning

Hackathons, workshops and seminars often see sudden participant surges beyond initial estimates.



## Conflicts go undetected

Manual coordination leads to double-booked rooms and scheduling clashes.



## Space is used inefficiently

Without occupancy tracking, capacity mismatches and overcrowding are common.

# Existing System

## How rooms & events are managed today

- Room status tracked informally — notice boards, spreadsheets, or word of mouth
- Event coordination happens over calls, messages and emails between organizers and admin staff
- No shared calendar view of bookings across departments
- Attendance and occupancy are estimated only after an event, if at all

## LIMITATIONS



### No single source of truth

Every stakeholder checks a different, often outdated, source.



### Reactive, not proactive

Conflicts and overcrowding are discovered only after they occur.



### Slow decision-making

Admins cannot quickly compare expected vs actual demand.



### Not scalable

Manual coordination breaks down as the number of events grows.

# Proposed System

A centralized, web-based platform where students discover rooms and events, and administrators allocate space, monitor registrations, and track real occupancy — all from one dashboard.



## Live Room Availability

View real-time room status, capacity and schedules before booking.



## Event Scheduling

Admins create and manage events, allocate rooms and set expected turnout.



## Participant Registration

Students register for events; the system tracks registrations against capacity.



## Conflict Detection

Automatically flags overlapping bookings and capacity mismatches.



## Admin Dashboard

A single console to monitor bookings, rooms and occupancy trends.



## Occupancy Tracking

Compares expected vs. actual attendance to guide future planning.

# Objectives

- 1 Provide a centralized platform for room and event management across campus
- 2 Give students real-time visibility into room availability and schedules
- 3 Streamline event creation, room allocation and participant registration
- 4 Detect and prevent room-booking conflicts before they occur
- 5 Track expected vs. actual occupancy to support data-driven decisions
- 6 Improve overall utilization of campus spaces and reduce manual effort

# Scope

## IN SCOPE

- ✓ Room catalog with capacity, building & equipment details
- ✓ Event creation, scheduling and room allocation by admins
- ✓ Student registration for events with live seat tracking
- ✓ Booking conflict detection across rooms and time slots
- ✓ Occupancy dashboard comparing expected vs. actual turnout
- ✓ Role-based access for students and administrators

## OUT OF SCOPE (for now)

- Online payment or fee collection for events
- Native mobile applications (web is responsive instead)
- Automated push notifications / SMS alerts
- Integration with external college ERP systems
- Advanced analytics or predictive forecasting
- Multi-campus / multi-institution support

# Technology Identification



## React.js

Frontend framework for the web interface



## Node.js

Backend JavaScript runtime environment



## Express.js

Backend framework for REST APIs



## MySQL

Relational database for system data



## MySQL Workbench

Database design & SQL development



## Postman

Testing and debugging REST APIs



## Visual Studio Code

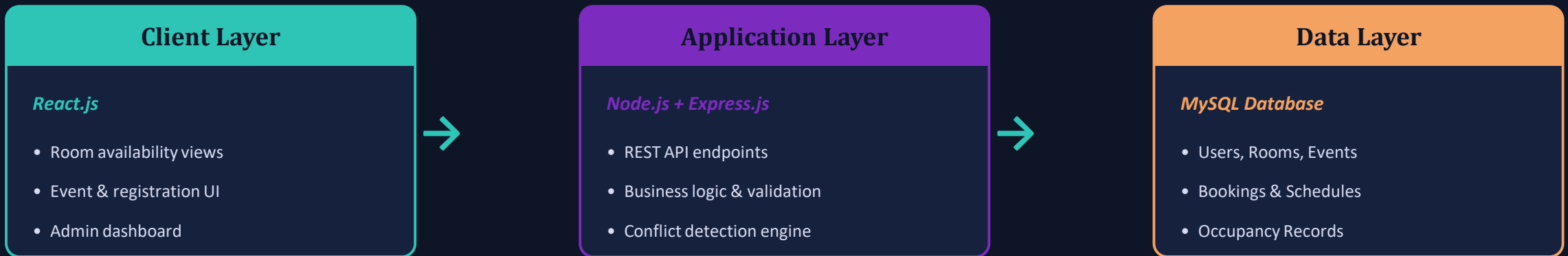
Primary development environment



## Git & GitHub

Version control & source management

# Basic Architecture



*Client requests flow through the Express REST API, which applies business rules (e.g. conflict checks) before reading or writing to MySQL — keeping presentation, logic and data cleanly separated.*

# Team Details

## TEAM 1 · SECTION 10



**J. Geya Reddy**

Roll No. 2520090123



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## GUIDE

**Divya Priya Degala**

## COURSE CONTEXT

### COURSE

25CS1302E — Database Systems Engineering & Distributed Backend Development

### TERM

Trimester 4, 2026–2027

### PROJECT

Event & Room Occupancy Management System

### STACK

React · Node.js · Express.js · MySQL

# Abstract Summary

The Event & Room Occupancy Management System replaces informal, manual coordination with a single web platform. Students get a clear view of room availability and can register for events confidently; administrators gain the tools to allocate rooms, resolve conflicts early, and track how spaces are actually used — turning guesswork into data-backed decisions for campus resource planning.

**3-Tier**

Architecture

**6+**

Core Entities

**3NF**

Normalized Schema

**2**

User Roles

*Abstract submitted · Ready for Review 1 evaluation*